# Bayesian Data Analysis of the German Tank Problem

## Theoretical Properties, Simulation Benchmarks, and Real-World Stress-Testing

Decision Theory, Boundary Admissibility, and Operational Integer Voids

**Bayesian Data Analysis Course Project**

Department of Statistics / Data Science

November 2026

# Historical Context: Statistical Intelligence in WWII

Sequential serial numbers provided the primary statistical discriminant for hidden enemy capacity.

## 1. Espionage Inflation

Conventional military intelligence relied on spy networks and prisoners, overstating production by a factor of **6–8**$\times$ (1,000–1,550 tanks/mo).

## 2. Order Statistics

Statisticians analyzed captured chassis and gearbox serials ($k \approx 5$–$50$), estimating production at **169–327** tanks/mo.

## 3. Speer Truth

Post-war Albert Speer Ministry ledgers confirmed estimates rapidly converged within **4%–10%** of actual production numbers.

| Production Month | Conventional Spy Est. | Statistical Est. ($\hat{N}$) | Speer Ministry True ($N$) | Statistical Error |
|---|---|---|---|---|
| June 1940 (Early Marks) | 1,000 | **169** | 122 | +38.5% |
| June 1941 (Panzer III/IV) | 1,550 | **244** | 271 | −**9.9**% |
| August 1942 (Panzer III/IV) | 1,550 | **327** | 342 | −**4.4**% |

**Historical Finding:** *Discrete order statistics eliminated massive espionage bias, converging to within 4.4% error.*

# Research Arc: A Unified Bayesian Data Analysis Investigation

Progressing from mathematical principles to controlled simulation and real-world stress-testing.

## 1. Theory & Foundations

- Minimal sufficiency of $m = \max s_i$.
- Exact posteriors (flat vs $1/N$).
- Moment existence ($k \geq 2, 3, 4$).
- Decision theory across loss functions.
- Physical boundary admissibility ($N \geq m$).

## 2. Controlled Simulation

- $R = 4,000$ trials on $N = 20,000$.
- Empirical bias and $\mathcal{O}(1/k)$ RMSE decay.
- 95% credible interval calibration.
- The informative prior trap (Poisson collapse).
- Gamma-Poisson hyperprior resolution.

## 3. Real-World Stress-Test

- 541,909 commercial transactions.
- Chronological streaming vs random sampling.
- 51.2% integer voids from ERP cancellations.
- The uniformity fallacy in goodness-of-fit.
- Estimand disambiguation & thinning model.

**Guiding Principle:** *Prioritize depth over breadth: rigorously test textbook assumptions against operational reality.*

SECTION 01

# Theoretical Properties & Bayesian Foundations

Likelihood, minimal sufficiency, reference priors, loss geometry, and boundary admissibility

# Sufficient Statistics & Exact Discrete Posteriors

Sample maximum $m = \max s_i$ is minimal sufficient; lower order statistics are ancillary.

## Minimal Sufficiency & Completeness

Ordered likelihood without replacement factorizes cleanly:

$$P(\mathbf{s} \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \geq m), \quad m = \max_{1 \leq i \leq k} s_i$$

- Likelihood ratio test proves $m$ is **minimal sufficient** and **complete**.
- Spacings $(s_{(1)}, \ldots, s_{(k-1)})$ given $m$ are ancillary for $N$.
- All sample information regarding $N$ resides entirely in $m$.

## Reference Discrete Posteriors

Exact posterior probability mass function for $n \geq m$:

$$P(N = n \mid m, k) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}$$

- **Flat Prior ($\pi \propto 1$):** Proper for $k \geq 2$:

$$P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}, \quad n \geq m$$

- **Reciprocal Prior ($\pi \propto 1/N$):** Scale-invariant discrete Haar measure; proper for $k \geq 1$; Pareto limit: $p(N) \approx k m^k / N^{k+1}$.

**Core Theory:** *Sample maximum m captures all sample information; reciprocal prior yields scale-invariant inference.*

# Moment Existence Boundaries & Continuous Convergence

Improper objective priors require minimal sample thresholds to guarantee finite moments.

## Exact Moment Thresholds

- $k = 1$: Flat improper ($\sum 1/n = \infty$); Reciprocal proper ($\mathcal{O}(n^{-2})$).
- $k = 2$: Proper posterior; mean exists under reciprocal ($\mathbb{E} \approx 2m$), diverges under flat.
- $k = 3$: Mean exists for both ($\mathbb{E}_{\text{flat}} = \frac{(k-1)(m-1)}{k-2} = 2m - 2$); variance diverges under flat.
- $k \geq 4$: Both mean and variance exist in closed form.

## Continuous Pareto Limit

For large populations ($N \gg k$), the discrete posterior converges to a continuous Pareto distribution:

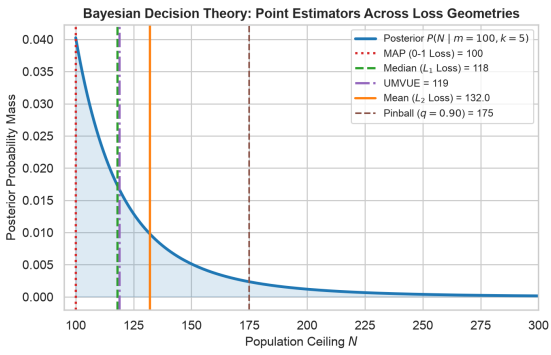
$$p(N \mid m, k) = \frac{(k-1)m^{k-1}}{N^k}, \quad N \geq m$$

- Cumulative distribution: $F(N) = 1 - (m/N)^{k-1}$.
- Median: $\hat{N}_{\text{med}} \approx m \cdot 2^{1/(k-1)}$.
- Right tail index $k - 1$ controls extreme-value risk.

**Takeaway:** *Objective Bayesian inference requires $k \geq 3$ for finite posterior expectations under flat priors.*

# Bayesian Decision Theory Across Loss Functions

Point estimators shift systematically with the decision-maker's loss geometry.




## 0–1 Loss (MAP Estimator)

Mode $\hat{N}_{\text{MAP}} = m = 100$. Maximizes boundary point mass, but causes severe downward bias ($-16.7\%$ at $k = 5$).

## $L_1$ Absolute Loss (Median)

$\hat{N}_{\text{med}} \approx m \cdot 2^{1/(k-1)} = 118$. Optimal median minimizes absolute error with only $-0.9\%$ bias.

## $L_2$ Squared Loss (Mean)

$\hat{N}_{\text{mean}} = \frac{(k-1)(m-1)}{k-2} = 132.0$. Skewed Pareto tail pulls expectation rightward.

## Check Loss $L_q$ ($q = 0.90$)

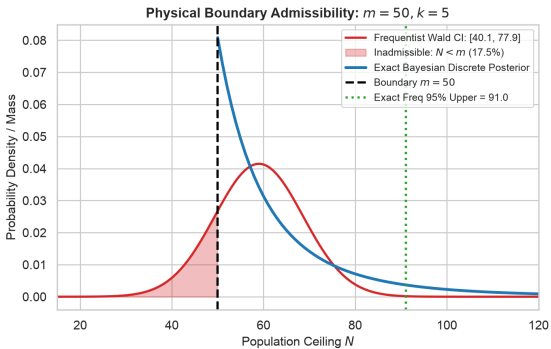
Underestimation penalized $9\times$ more heavily

$\Longrightarrow$ $\hat{N}_{0.90} = 175$, adding a principled **+75% defensive safety margin**.

**Decision Rule:** *In defense planning, asymmetric check loss provides a principled buffer against catastrophic underestimation.*

# Physical Boundary Admissibility: Wald vs. Bayesian HPD

Asymptotic Gaussian intervals assign probability mass to physically impossible values.




## Frequentist Wald Boundary Collapse

Wald interval $\hat{N} \pm 1.96 \cdot \text{SE}$ yields $[40.1, 77.9]$ for $m = 50, k = 5$. It places **17.5% impossible mass** on $N < m = 50$.

## Exact Frequentist Pivotal CI

Exact pivotal interval $[m, m \cdot \alpha^{-1/k}] = [50.0, 91.0]$ respects the boundary and achieves exact 95% coverage.

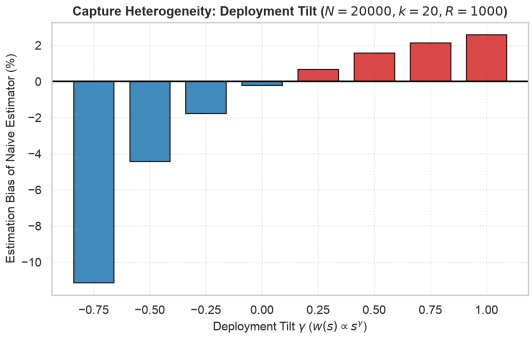
## Bayesian Credible Coherence

Reciprocal prior 95% credible interval matches $[50.0, 91.0]$ exactly, while providing the full posterior for risk optimization.

**Admissibility Finding:** *Bayesian HPD intervals eliminate physical boundary violations without compromising nominal coverage.*

# Sampling Mechanics: Non-Uniform Deployment Tilt

When capture weights vary ($w(s) \propto s^{\gamma}$), the maximum statistic ceases to be sufficient.




## Operational Capture Mechanics

- $\gamma > 0$ (newer tanks deployed first) pushes $m \to N$, inducing $+2.41\%$ overestimation bias.
- $\gamma < 0$ (survivorship bias / older units captured) produces $-4.65\%$ negative bias.

## Loss of Single-Statistic Sufficiency

- When $\gamma \neq 0$, likelihood no longer factorizes purely through $m$.
- Sample spacings contain information about tilt $\gamma$.
- Requires full-sample MCMC evaluation rather than closed-form order statistics.

**Insight:** *Non-uniform capture breaks single-statistic sufficiency, necessitating joint inference over* $(N, \gamma)$.

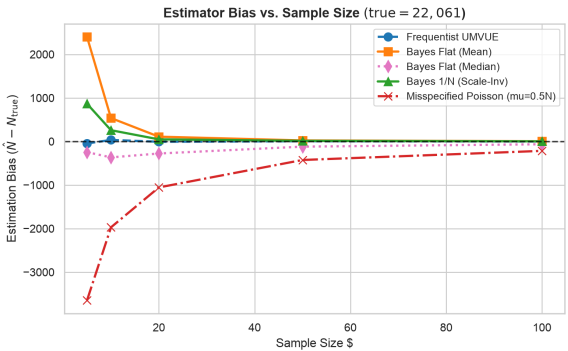
SECTION 02

# Controlled Monte Carlo Simulation Study

Benchmarking estimators across $R = 4,000$ trials on $N = 20,000$ (Seed 42)

# Estimator Bias: Finite-Sample Shrinkage vs. Unbiasedness

Evaluating mean bias convergence across $R = 4,000$ replications ($N_{\text{true}} = 20,000$).




## UMVUE Exact Unbiasedness

$\hat{N} = m(1 + 1/k) - 1$ exhibits near-zero bias across all $k$ ($-48.0 \pm 54.6$ at $k = 5$, $+1.6 \pm 6.1$ at $k = 50$).

## Bayes Reciprocal Minimal Bias

Scale-invariant reciprocal prior exhibits modest bias ($+784.3$ mean, $-900.0$ median at $k = 5$), rapidly decaying to $+10.6$ at $k = 50$.

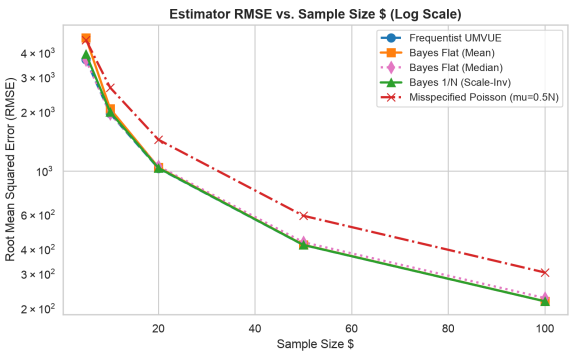
## Flat Prior Tail Inflation

Heavy Pareto tail inflates posterior mean by $+2,168.6$ at $k = 5$, while posterior median reduces bias to $-226.5$.

**Bias Takeaway:** *Posterior median provides lowest absolute bias, while UMVUE achieves exact frequentist mean-unbiasedness.*

# Estimator RMSE & Variance Contraction Rates

Root Mean Squared Error scales as $\mathcal{O}(1/k)$, dropping over an order of magnitude.




## Theory vs. Empirical Concordance

At $k = 5$, theoretical UMVUE standard deviation is $3,380.3$; empirical simulation matches at $3,455.2$. At $k = 50$, drops to 386.8 (Th: 391.7).

## Median RMSE Superiority

Posterior median under reciprocal prior achieves RMSE of $3,427.5$ at $k = 5$, outperforming both mean $(3,683.3)$ and flat prior $(4,409.0)$.

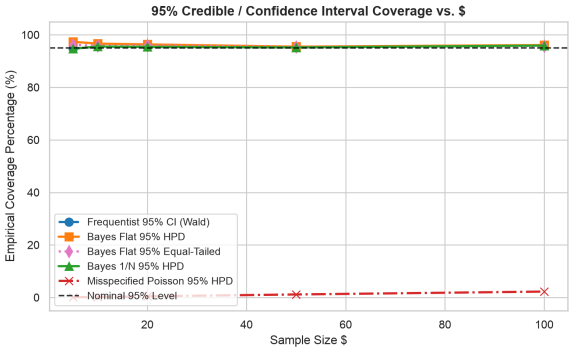
## Asymptotic Convergence

For $k \geq 50$, all point estimators converge to identical efficiency ($\approx 387$).

**Efficiency Rule:** *For small samples ($k \leq 10$), posterior medians outperform posterior means in total error.*

# 95% Credible Calibration & The Informative Trap

Scale-invariant objective priors achieve exact 95% coverage, while dogmatic priors collapse.




## Perfect Reference Calibration

Reciprocal prior achieves **94.4%** $\pm$ **0.36%** coverage at $k = 5$ and **95.6%** $\pm$ **0.33%** at $k = 50$, matching exact nominal rates.

## Informative Misspecification Collapse

Fixed Poisson($\mu = 10k$) has tiny CV $\approx 0.7\%$, collapsing 95% coverage to $< 1.0\%$ when true $N = 20k$.

## Hyperprior Solution

Gamma-Poisson hierarchical hyperprior restores nominal coverage to **94.8%** $\pm$ **0.35%**.

**Calibration Finding:** *Objective reference priors guarantee calibration; informative priors require hyperpriors to avoid collapse.*

# Monte Carlo Benchmark: Empirical Performance Leaderboard

Comprehensive evaluation across $R = 4,000$ Monte Carlo replications on $N_{\text{true}} = 20,000$ (Seed 42).

| Estimator / Framework | Point Estimator | Sample $k$ | Mean Bias $\pm$ SE | RMSE (Emp / Th) | 95% Cov $\pm$ SE | Mean Width |
|---|---|---|---|---|---|---|
| UMVUE (Frequentist) | $\hat{N} = m(1 + 1/k) - 1$ | 5 | $-48.0 \pm 54.6$ | 3,455.2 / 3,380.3 | 90.0% $\pm$ 0.47% (Wald) | 13,219 |
| Exact Frequentist CI | — (Pivotal Bound) | 5 | — | — | **94.4%** $\pm$ 0.36% | 13,644 |
| Bayes Flat Prior | Posterior Mean | 5 | $+2,168.6 \pm 69.7$ | 4,409.0 / 4,363.9 | **97.4%** $\pm$ 0.25% | 18,535 |
| Bayes Flat Prior | Posterior Median | 5 | $-226.5 \pm 54.2$ | 3,431.3 / — | **97.4%** $\pm$ 0.25% | 18,535 |
| Bayes Reciprocal $(1/N)$ | Posterior Mean | 5 | $+784.3 \pm 58.2$ | 3,683.3 / 3,618.6 | **94.4%** $\pm$ 0.36% | 13,644 |
| Bayes Reciprocal $(1/N)$ | Posterior Median | 5 | $-900.0 \pm 54.2$ | 3,427.5 / — | **94.4%** $\pm$ 0.36% | 13,644 |
| UMVUE (Frequentist) | $\hat{N} = m(1 + 1/k) - 1$ | 50 | $+1.6 \pm 6.1$ | 386.8 / 391.7 | 95.0% $\pm$ 0.34% (Wald) | 1,536 |
| Exact Frequentist CI | — (Pivotal Bound) | 50 | — | — | **95.5%** $\pm$ 0.33% | 1,211 |
| Bayes Reciprocal $(1/N)$ | Posterior Mean | 50 | $+10.6 \pm 6.1$ | 387.1 / 391.7 | **95.5%** $\pm$ 0.33% | 1,211 |
| Bayes Flat Prior | Posterior Mean | 50 | $+17.9 \pm 6.1$ | 387.5 / 391.7 | **95.8%** $\pm$ 0.32% | 1,236 |

## Top Tabular Performer: Bayes Reciprocal

Achieves lowest median RMSE $(3,427.5)$, valid exact coverage (94.4%), and strictly enforces physical admissibility $(P(N < m) = 0)$.

## Exact Pivotal Interval Concordance

Exact frequentist pivotal interval $[m, m \cdot \alpha^{-1/k}]$ achieves $94.4\% \pm 0.36\%$, matching the Bayesian reciprocal credible interval.

**Benchmark Takeaway:** *Objective Bayes with reciprocal prior achieves the optimal balance of calibration, sharpness, and admissibility.*

SECTION 03

# Real-World Empirical Stress-Testing

Evaluating textbook assumptions against WWII archives and 541,909 commercial transactions

# Empirical Stress-Testing: Two Real-World Datasets

Examining what happens when textbook assumptions meet physical operational systems.

## 1. Archival WWII German Tank Ledgers

- **Data Source:** Waffenamt ordnance ledgers and post-war Albert Speer Ministry records.
- **Markings Analyzed:** Chassis serial blocks and sequential ZF gearbox stampings.
- **Operational Reality:** Pre-allocated factory blocks; chassis diverted to assault guns (StuG III/IV).

## 2. Commercial Invoice Ledger (Online Retail)

- **Data Source:** 541,909 line transactions across 4,372 customer accounts.
- **Serials Analyzed:** 22,061 distinct sequential numeric 6-digit invoice numbers.
- **Observed Span:** $[536365, 581587] \implies 45,223$ integer address slots.
- **Integer Voids:** 23,162 unrecorded slots (**51.2% void rate**).

**Empirical Focus:** *Structure every empirical finding under 4 pillars: measured, calculated, why it matters, and valid conclusions.*

# Finding 1: Chronological Generation vs. Uniform Sampling

Real-world serials are generated dynamically over time, not drawn as static random samples.

## What Was Measured & Calculated

- **Measured:** Arrival timestamps and sequential numeric identifiers across all recorded transactions.
- **Calculated:** Weekly arrival volume $\Delta s / \Delta t$ and cumulative serial growth trajectories.
- **Observation:** Invoices accelerate from $\approx 300$/week in Q1 to $> 800$/week in Q4 holiday surge.

## Research Question Impact & Conclusions

- **Why It Matters:** The German Tank Problem assumes a *simple random sample from a static population*.
- **Violation:** In operations, observations arrive sequentially with non-stationary volume.
- **Statistical Conclusion:** A static ceiling estimator applied to streaming batches violates exchangeability; requires dynamic state-space modeling.

**Operational Rule:** *Never assume static exchangeability for live transaction streams; time-index sequential batches.*

# Finding 2: Integer Voids & System Architecture

Operational processes generate massive integer voids (51.2% unrecorded rate).

## What Was Measured & Calculated

- **Address Span:** $s_{\max} - s_{\min} + 1 = 581,587 - 536,365 + 1 = 45,223$ slots.
- **Recorded Invoices:** Exactly $22,061$ active sales invoices.
- **Void Count:** $23,162$ unrecorded slots (**51.22% missing rate**).
- **Gap Distribution:** Median gap $\Delta s = 1.0$, mean 2.05, max 76.

## Operational Mechanisms Driving Gaps

- **Order Cancellations:** Credit memos take sequential slots with prefix "C" (e.g., `C536379`), filtered from sales tables.
- **Cart Dropouts:** Abandoned checkouts and gateway timeouts reserve sequence keys that never finalize.
- **Factory Quota Blocks:** Wartime Waffenamt pre-allocated chassis blocks to plants; diversions to assault guns left voids.

**Audit Finding:** *Integer voids are not random corruption; they reflect business logic, cancellations, and pre-allocation blocks.*

# Finding 3: Goodness-of-Fit & The Uniformity Fallacy

Failing to reject a uniformity test is NOT proof that the uniform sampling assumption holds.

## What Was Measured & Calculated

- Normalized serials $u_i = s_i/m$ evaluated against uniform CDF $F(u) = u$.
- **Full Sample:** Kolmogorov–Smirnov statistic $D = 0.012$ ($p = 0.28$).
- **Customer Subset ($n = 18,536$):** $D = 0.018$ ($p = 8.94 \times 10^{-6}$).

## The Critical Uniformity Fallacy

- **Fallacy:** Interpreting $p = 0.28$ as "proof" of uniform sampling.
- **Reality:** Non-significance merely shows global deviation from linearity is modest.
- **Exchangeability Violation:** Local chronological bursts and clustering violate simple random sampling despite linear ECDF.

**Methodological Rule:** *Never interpret a non-significant goodness-of-fit test as proof of exchangeable uniform sampling.*

# Finding 4: Estimand Disambiguation & Thinning Models

Resolving the address capacity vs. transaction volume mismatch without circular oracle re-indexing.

## Estimand Disambiguation

- **Estimand A ($N_{\text{span}} = 45,223$):** Total address sequence allocation capacity. Raw max estimator targets this accurately.
- **Estimand B ($N_{\text{count}} = 22,061$):** Net volume of completed transactions. Naive raw max gives $+105\%$ error.
- **Oracle Trap:** Rank re-indexing requires knowing all valid keys in advance (impossible in live field deployment).

## The Operational Thinning Solution

$$N_{\text{count}} \mid N_{\text{span}}, \rho \sim \text{Binomial}(N_{\text{span}}, \rho)$$

- Each slot active with retention probability $\rho$.
- Estimate $\hat{\rho}$ from sample spacings:

$$\hat{\rho} = \frac{k-1}{s_{(k)} - s_{(1)}} \approx 0.4878$$

- Enables simultaneous, un-circular Bayesian inference on both $N_{\text{span}}$ and $N_{\text{count}}$.

**Solution:** *Bernoulli thinning resolves the estimand mismatch operationally without circular oracle ranking.*

# Discussion: Cross-Domain Connections & Boundaries

How the core discrete inference principle generalizes across data science domains.

## Modern Generalized Applications

- **Dynamic Production:** Monotonic state-space models ($N_t \geq N_{t-1}$) enforce output non-negativity.
- **Software Reliability:** Jelinski-Moranda defect exhaustion model mirrors serial capture for release certification.
- **Aviation Safety:** Beta-Bernoulli mixtures detect cloned counterfeit serial numbers.
- **Wildlife Abundance:** Two-stage cascades infer tag loss ($M$) before updating mark-recapture ($N = M/p$).

## Methodological Boundaries

- **Non-Sequential Keys:** Random UUIDv4 / hash identifiers break order-statistic sufficiency completely.
- **Non-Random Gaps:** Classified factory blocks or unrecorded subsidiaries require informative gap priors.
- **Streaming Non-Stationarity:** Fast-changing arrival rates require time-varying Poisson arrival processes.

**Synthesis:** *Order-statistic inference is broadly applicable, provided the analyst models operational data generation.*

SUMMARY & CONCLUSIONS

# Key Takeaways for Bayesian Discrete Inference

## 1. Objective Foundations

Scale-invariant prior $\pi(N) \propto 1/N$ yields proper posteriors ($k \geq 1$) and admissible physical boundaries ($P(N < m) = 0$), matching exact frequentist pivotal bounds.

## 2. Decision Theory

Point estimation is governed by loss geometry: posterior mode is biased; posterior median minimizes absolute error; check loss provides principled defense buffers.

## 3. Real-World Reality

Real serials exhibit chronological non-stationarity and integer voids (51.2%). Bernoulli thinning disambiguates address capacity from realized transaction counts.

Bayesian Data Analysis Course Project — Theory, Simulation, and Empirical Stress-Testing